Parth Sachin Deshmukh

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Education

Texas A&M University, College Station, TX Master of Science in Computer Science — May 2027
Coursework: Artificial Intelligence, Analysis of Algorithms, Computer-Human Interaction, Data Science

Vishwakarma Institute of Information Technology, Pune, India Bachelor of Technology in Information Technology — GPA: 9.12 / 10.00 — May 2024

Skills

Languages: Python, SQL, C++, JavaScript
Key Technologies: REST APIs, Git/GitHub, Docker, AWS (EC2, S3), PostgreSQL, MySQL
Specializations: Data Structures & Algorithms, System Design, Object-Oriented Programming, Agile/Scrum
Additional: React.js, Node.js, PHP, Machine Learning, Neural Networks, OpenCV, JIRA, Problem-Solving, Debugging

Experience

Research Intern — Indian Institute of Technology (IIT) Bombay, Mumbai, India — Feb 2024 – Sept 2024
- Developed a **PHP and JavaScript–based** data visualization and theme builder using the **Yii Framework**, enabling 20+ professors to design and manage analytics dashboards and custom layouts.
- Built a full-stack web application **Figma-to-HTML parser** with drag-and-drop functionality and a **reusable data reporting theme system** for 100+ reports, supporting structured data presentation and visualization.
- Implemented **dynamic, data-driven styling and automated visualization logic** to streamline analytics workflows.
- Optimized **MySQL database** queries with **indexing strategies** and integrated **AJAX** for asynchronous data fetching, managing 500+ layout configurations and reducing report generation time by 80% through performance optimization.

UI/UX Design Intern — HTS Tech Solutions, Pune, India — Jun 2022 – Dec 2022
- Designed **interactive data visualization** dashboards for analytics applications, applying **UI/UX design** principles.
- Conducted stakeholder feedback sessions to identify key data reporting and visualization requirements.
- Created **wireframes** and **prototypes**, iterating based on feedback and ensuring **accessibility** standards compliance.

Projects

Restaurant Performance Analytics — Google BigQuery — Sept 2024
- Conducted data analysis on Grubhub and Uber Eats datasets using SQL query optimization and Google BigQuery, identifying delivery performance patterns and operational inefficiencies
- Built dashboards with KPI metrics and data visualization, applying statistical analysis to support decision making

ASL Detection Model — TensorFlow, Docker, AWS — July 2024
- Designed a deep learning model for **American Sign Language recognition** on labeled samples.
- Deployed the containerized model using **Docker on AWS EC2**, ensuring scalable and reproducible inference.

Superstore Sales Analysis — Power BI — August 2024
- Built **interactive Power BI dashboards** to visualize KPIs, profitability, and customer segmentation.
- Applied **DAX measures** and automated refresh pipelines to support real-time business analytics.

Devanagari Character Recognition — Python, OpenCV, TensorFlow — Capstone Project — 2024
- Processed **6,000+ handwritten character samples** using OpenCV and NumPy, engineered features for 46 character classes, and performed **data preprocessing** to improve model input.
- Applied **transfer learning with ResNet and VGG models**, fine-tuning them for improved character recognition.
- Built and evaluated a CNN classifier, achieving **90% accuracy**, covering **model development and evaluation**.

Achievements

- **Publications:** First author, Springer (Automated GUI Testing, UX); co-author, IEEE (Blockchain for Disaster Relief).
- **Leadership:** Led UI/UX team at OIR, VIIT, coordinating projects involving **data visualization and user analytics**.
- **International Exposure:** Completed Summer & Spring School on **Advanced Engineering and Data Science**, Tokushima University, Japan.